



1. Preview the iris table and output the first 5 rows (1 points)

code:

```
View(iris)
iris[1:5,]
```

Screenshot:

The screenshot shows the RStudio interface. The script editor contains the following code:

```
1 #Preview the iris table and output the first 5 rows (1 points)
2 #Print the numbers of rows and columns of the table (1 points)
3
4 View(iris)
5 iris[1:5,]
```

The console output shows the first 5 rows of the iris dataset:

```
> iris[1:5,]
  Sepal.Length Sepal.Width Petal.Length Petal.Width Species
1          5.1         3.5        1.4         0.2  setosa
2          4.9         3.0        1.4         0.2  setosa
3          4.7         3.2        1.3         0.2  setosa
4          4.6         3.1        1.5         0.2  setosa
5          5.0         3.6        1.4         0.2  setosa
```

The Environment pane on the right shows the loaded objects: patient_data (5 obs. of 3 variables), academic_plot (Large ggplot2::ggplot (16.9 MB)), expression_levels (num [1:3] 12.5 NA 8.3), feret_radius (num [1:121] 2.24 4.12 5.83 4.47 4.24 ...), feret_radius_clean (num [1:120] 2.24 4.12 5.83 4.47 4.24 ...), gene_names (chr [1:3] "TP53" "BRCA1" "EGFR"), math (num [1:5] 98 96 97 100 88), mean_log (Named num 1.75), mu (6.76412396694215), range_x (num [1:2] 1.41 21.47), and save_path_and_filename ("/Users/gaoyuchen/Desktop/Feret_Radius_Clean_Lognormal_Fit.tiff").

Conclusion:

Successfully output the first 5 rows

ID	Sepal.L	Sepal.W	Petal.L	Petal.W	Species
1	5.1	3.5	1.4	0.2	setosa
2	4.9	3.0	1.4	0.2	setosa
3	4.7	3.2	1.3	0.2	setosa
4	4.6	3.1	1.5	0.2	setosa

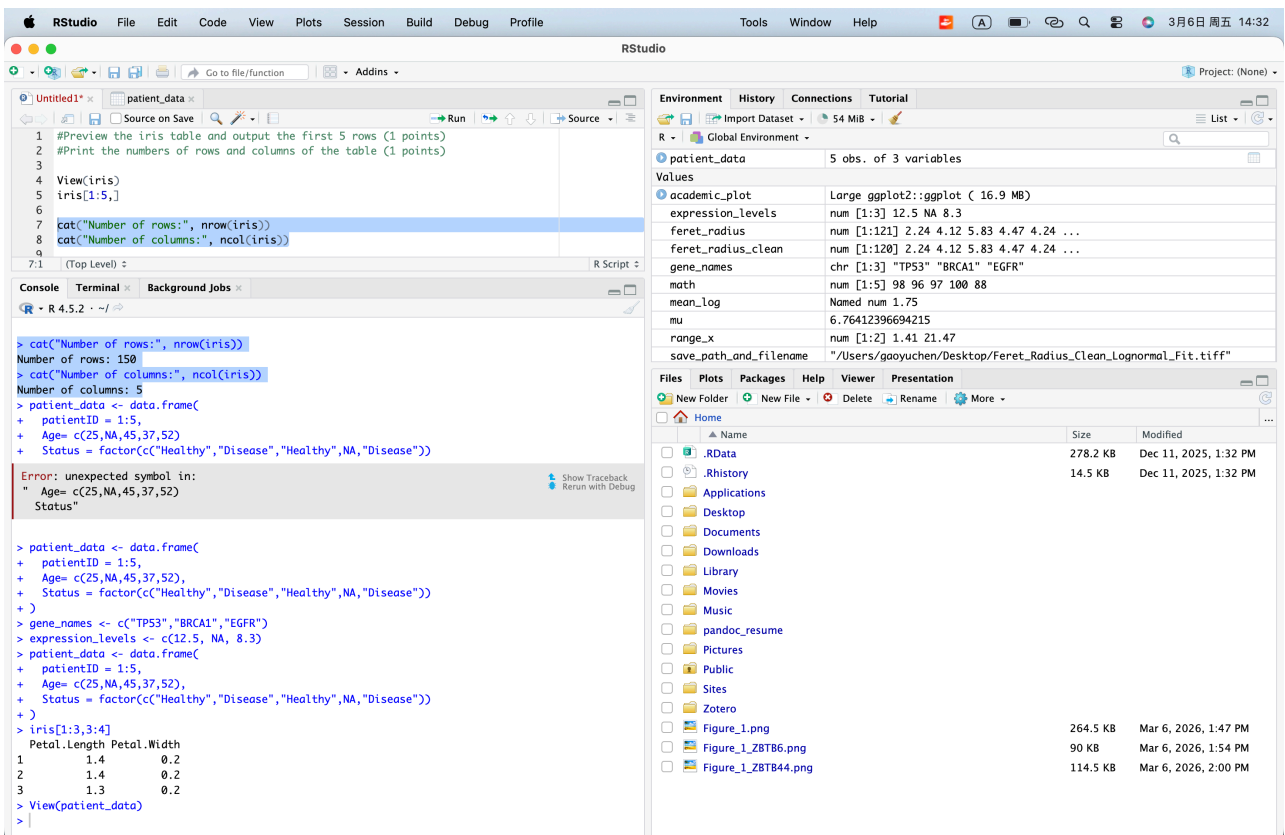
ID	Sepal.L	Sepal.W	Petal.L	Petal.W	Species
5	5.0	3.6	1.4	0.2	setosa

2. Print the numbers of rows and columns of the table

code:

```
cat("Number of rows:", nrow(iris))
cat("Number of columns:", ncol(iris))
```

Screenshot:



Conclusion:

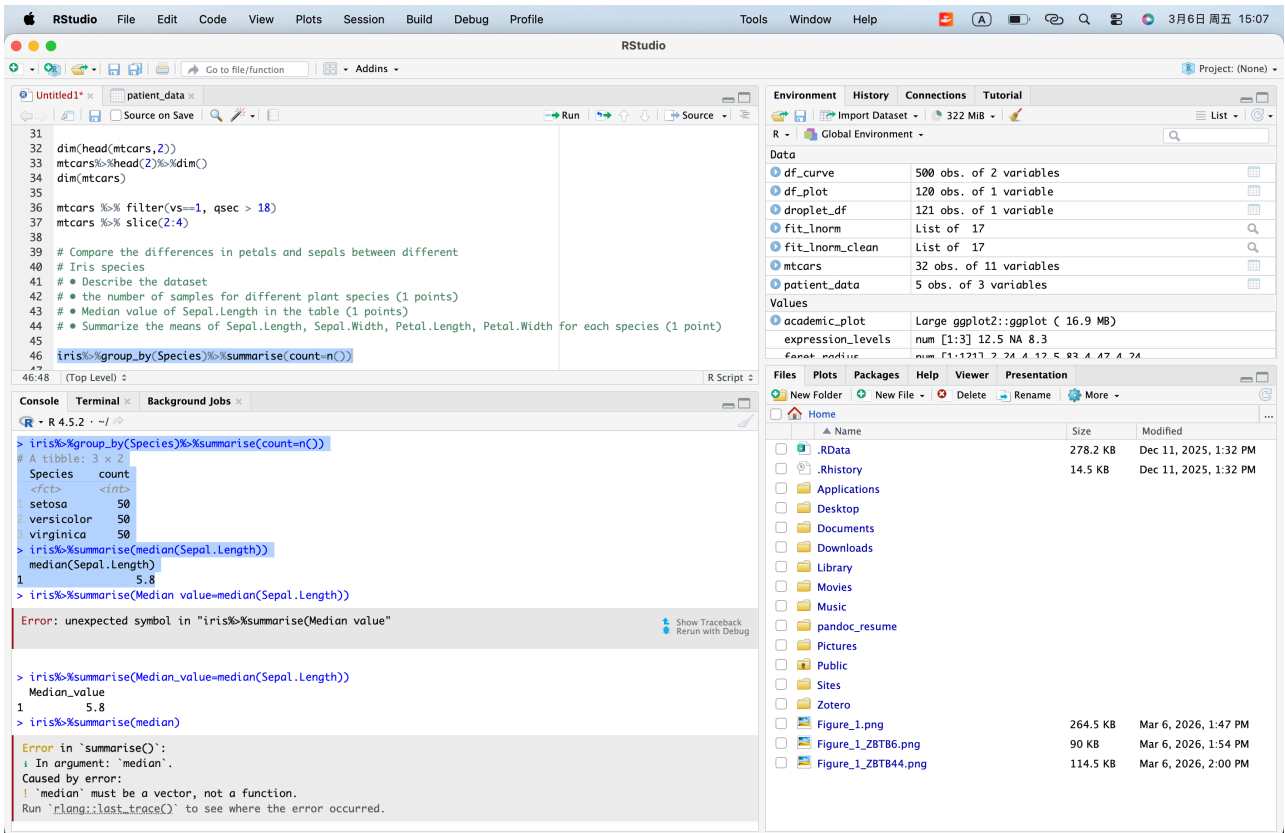
Successfully count the number of rows and columns of the table, there's 150 rows and 5 columns.

3. Describe the number of samples for different plant species (1 points)

code:

```
iris%>%group_by(Species)%>%summarise(count=n())
```

Screenshot:



Conclusion:

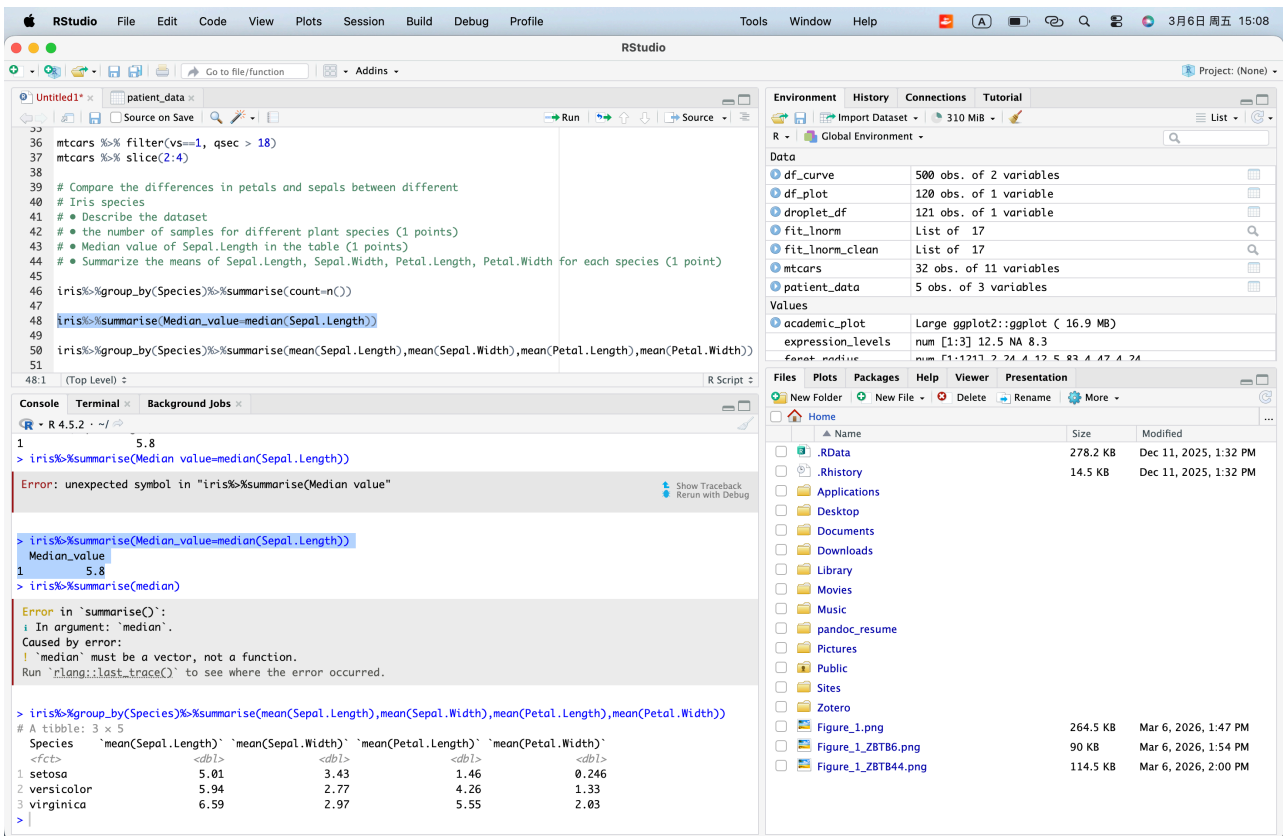
Successfully count the number of samples for different plant species, each species has 50 samples.

4. Median value of Sepal.Length in the table (1 points)

code:

```
iris%>%summarise(Median_value=median(Sepal.Length))
```

Screenshot:



Conclusion:

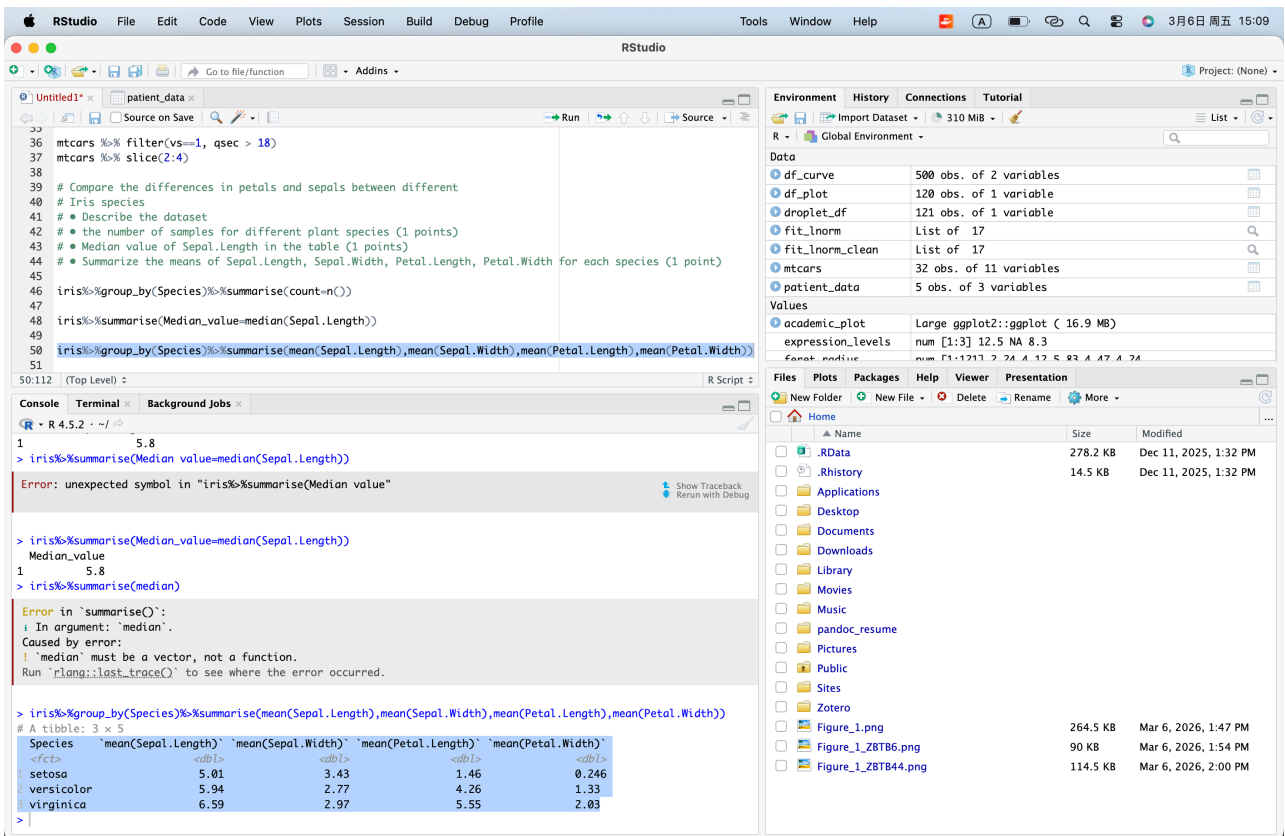
The Median value of `Sepal.Length` is 5.8.

- Summarize the means of `Sepal.Length`, `Sepal.Width`, `Petal.Length`, `Petal.Width` for each species (1 point)

code:

```
iris%>%group_by(Species)%>%summarise(mean(Sepal.Length),mean(Sepal.Width),mean(Petal.Length))
```

Screenshot:



Conclusion:

Successfully obtain the means of Sepal.Length, Sepal.Width, Petal.Length, Petal.Width for each species.

Species	m.Sepal.L	m.Sepal.W	m.Petal.L	m.Petal.W
setosa	5.01	3.43	1.46	0.246
versicolor	5.94	2.77	4.26	1.33
virginica	6.59	2.97	5.55	2.03

Add a column to the table annotating the main origin of each species (Tip: use full_join)

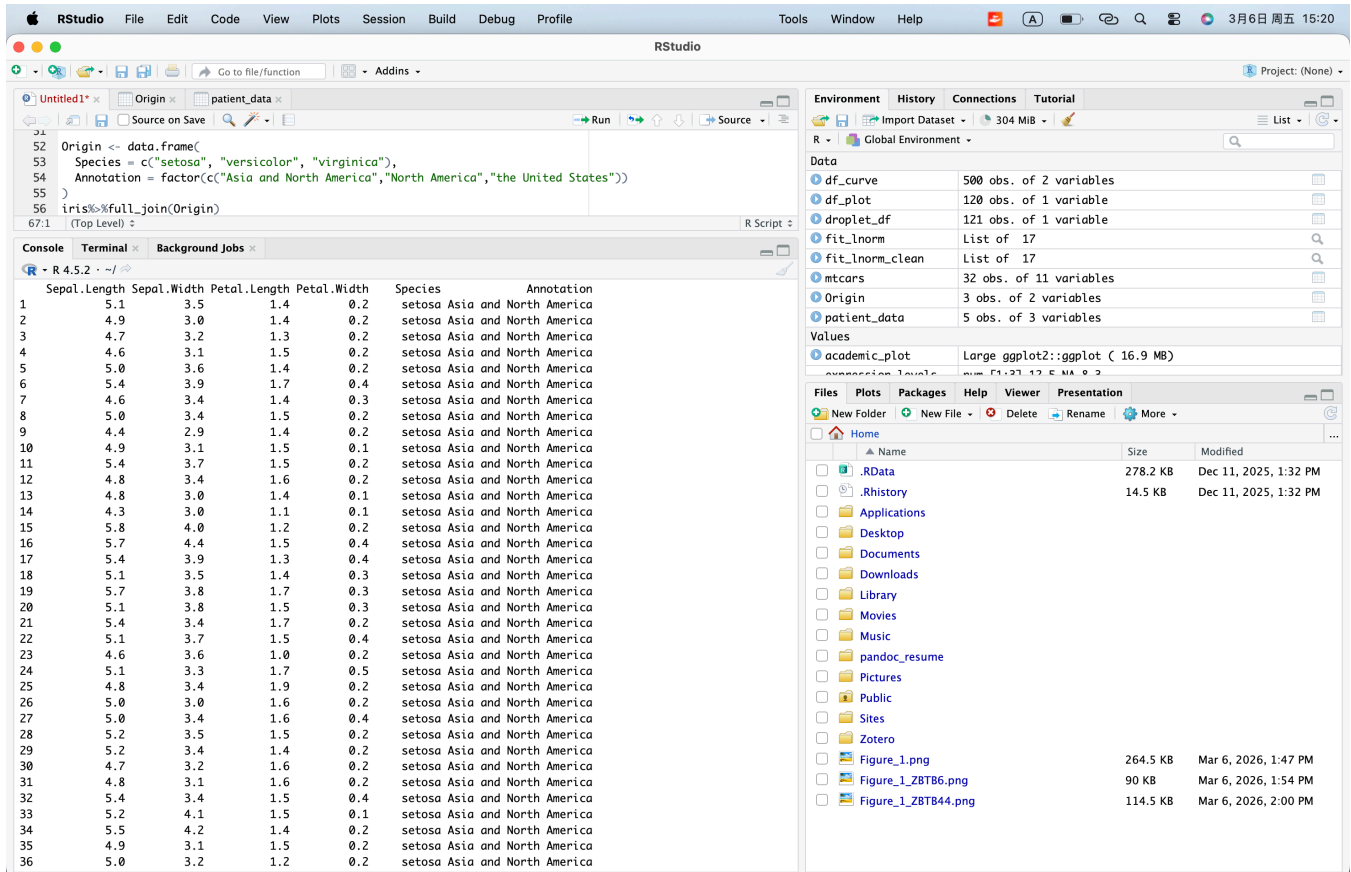
code:

```

Origin <- data.frame(
  Species = c("setosa", "versicolor", "virginica"),
  Annotation = factor(c("Asia and North America", "North America", "the United States"))
)
iris%>%full_join(Origin)

```

Screenshot:

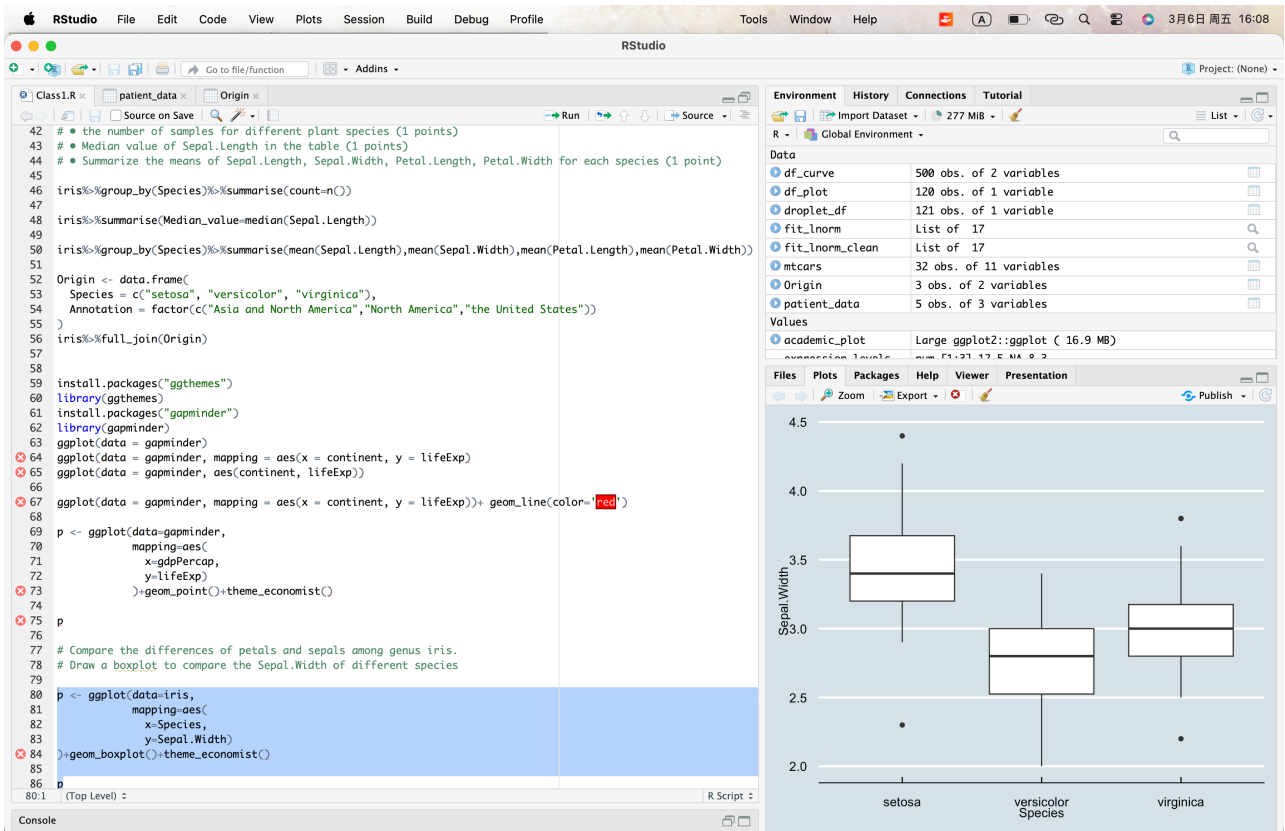


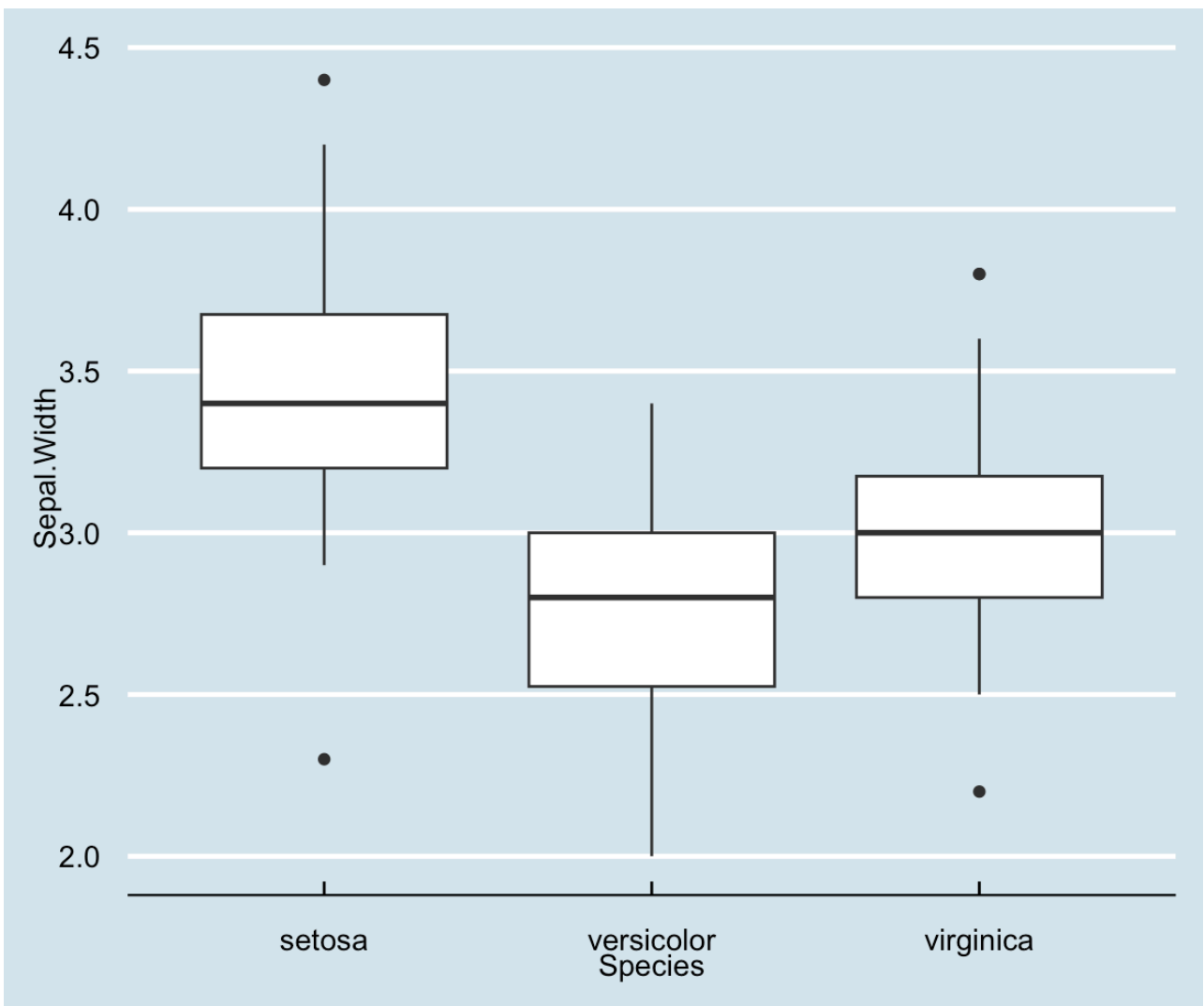
1. Draw a boxplot to compare the Sepal.Width of different species (2 point)

code:

```
p <- ggplot(data=iris,
            mapping=aes(
              x=Species,
              y=Sepal.Width)
            )+geom_boxplot()+theme_economist()
```

Screenshot:



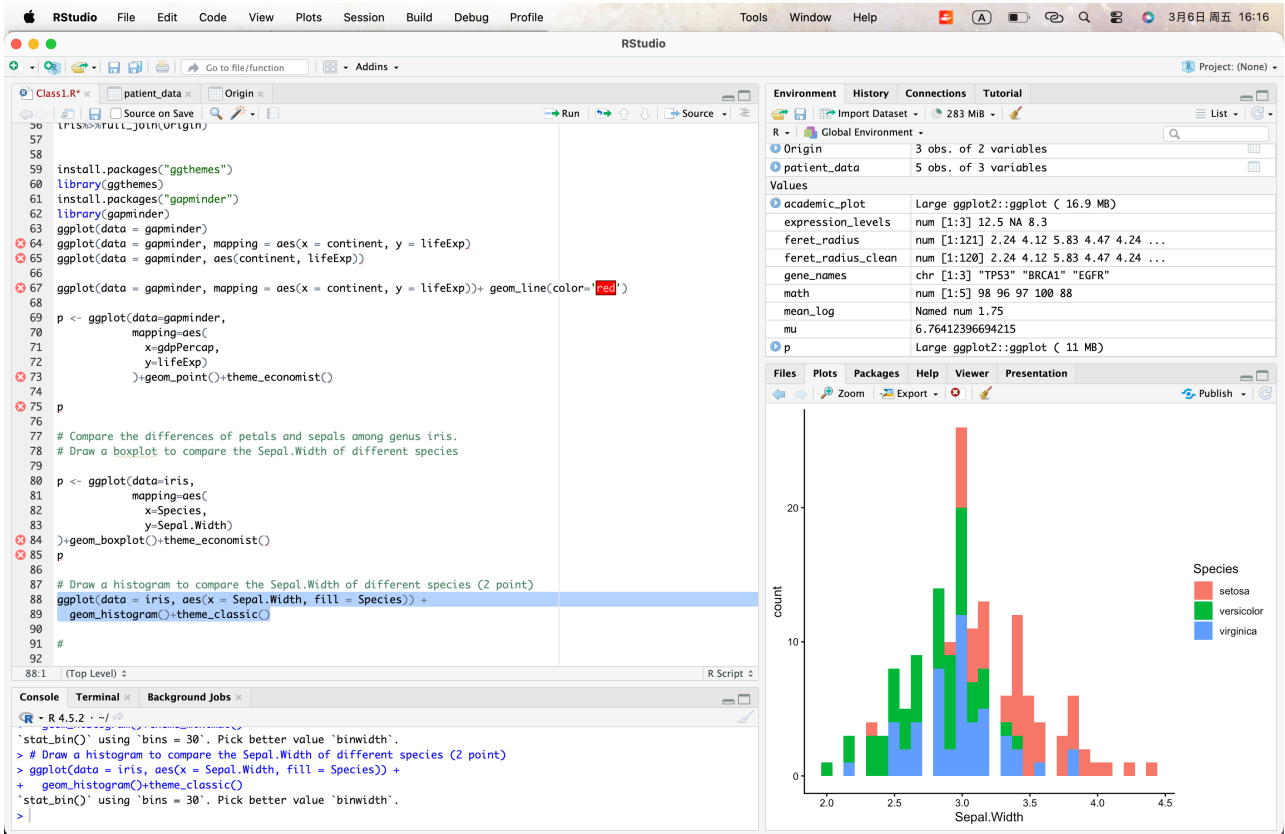


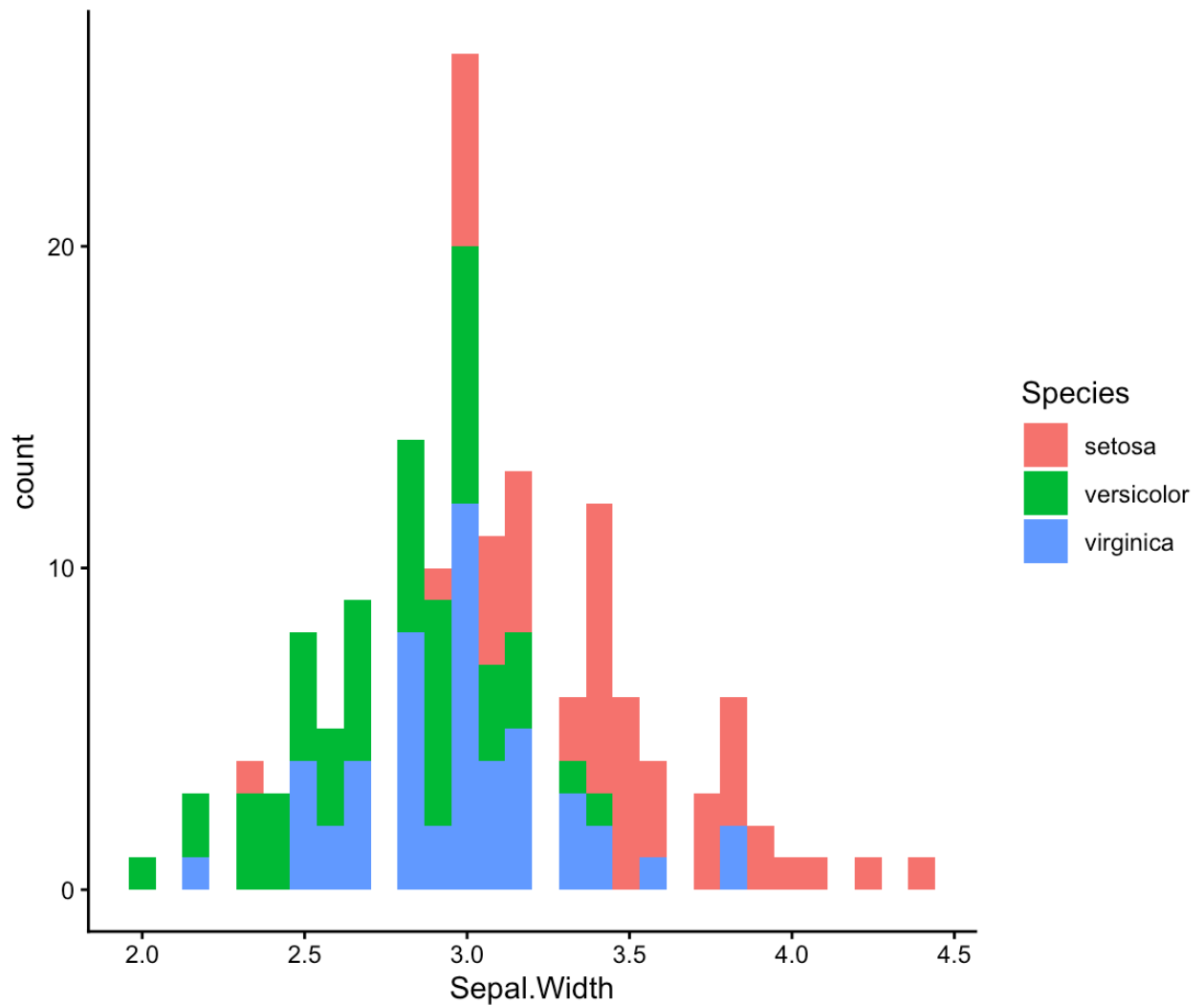
2. Draw a histogram to compare the Sepal.Width of different species (2 point)

code:

```
ggplot(data = iris, aes(x = Sepal.Width, fill = Species)) +geom_histogram()+theme_classic()
```

Screenshot:





3. Raise a scientific question about iris dataset (1 point)

To what extent can we mathematically define a "species boundary" using only floral measurements